

Pavel Zakharov

zakharov@math.uni-bielefeld.de | +49 15234632591 | Google Scholar

Education

Bielefeld University, Germany, Faculty of Mathematics, PhD Program	2026 – present
TU Dortmund, Germany, Department of Mathematics, PhD Program	2023 – 2026
Higher School of Economics, Moscow, Russia, Data Science, Math in Machine Learning, Master Program	2021 – 2023
Higher School of Economics, Moscow, Russia, Faculty of Computer Science, Applied Mathematics and Informatics, Bachelor Program	2017 – 2021
Physical-Mathematical School 2007, Moscow, Russia, Graduated with Golden Medal.	2017

Positions

Bielefeld University, Faculty of Mathematics, Research Assistant	2026 – present
TU Dortmund, Department of Computer Science, Research Assistant	2023 – 2026
University of New South Wales, Research Internship	2025
HSE University, Laboratory of Theoretical Computer Science, Research Assistant	2019 – 2023
Yandex LLC, Yandex.Routing, Yandex.Adtech, Software engineer	2019 – 2020

Research

Preprints

Fluctuations of the Ising free energy on Erdős-Rényi graphs, A. Coja-Oghlan, D. Kaaser, M. Rolvien, P. Zakharov, K. Zampetakis.	2026
The number of random 2-SAT solutions is asymptotically log-normal A. Chatterjee, A. Coja-Oghlan, N. Müller, C. Riddlesden, M. Rolvien, P. Zakharov, H. Zhu, To appear in <i>Theory of Computing</i> .	2024

Journal publications

A note on the maximum clique LP relaxation, P. Zakharov, <i>Operations Research Letters</i> 66 (2026) 107425.
The random k-SAT Gibbs uniqueness threshold revisited, A. Chatterjee, A. Coja-Oghlan, C. Greenhill, V. Pfenninger, M. Rolvien, P. Zakharov, K. Zampetakis, <i>Combinatorics, Probability and Computing</i> 35 (2026) 1–53.
WalkSAT is linear on random 2-SAT, P. Berenbrink, A. Coja-Oghlan, C. Cooper, T. Götze, L. Hintze, P. Zakharov, <i>SIAM Journal on Discrete Mathematics</i> 39 (4) (2025) 1939–1952.
Fractional colourings of random hypergraphs, D. Shabanov, P. Zakharov,

Russian Mathematical Surveys **78** (6) (2023) 183–184.

Asymptotic normality of the giant component size in a random bipartite graph,

D. Shabanov, P. Zakharov,

Trudy MIPT **15** (2023) 23–32 (in Russian).

On the maximal cut in a random hypergraph,

D. Shabanov, P. Zakharov,

Doklady Mathematics **104** (2021) 336–339.

Talks

31st British Combinatorial Conference,

Cardiff, United Kingdom.

Randomness in Computing,

Sheffield, United Kingdom.

Athens-Dortmund-Hamburg Workshop on Graphs, Networks, Optimization,

and Statistical Inference,

Corfu, Greece.

Midwinter meeting in Discrete Probability,

Umeå, Sweden.

Algorithmic Decision Theory Reading Group,

Sydney, Australia.

22nd International Conference on Random Structures and Algorithms,

Vienna, Austria.

28th International Conference on Randomization and Computation,

London, United Kingdom.

Algorithms, Dynamics, and Information Flow in Networks Seminar,

Online.

European Conference on Combinatorics, Graph Theory and Applications,

Online.

Research grants

Co-investigator on “Extremal properties of random graphs”. Russian Foundation for Basic Research and Iran National Science Foundation, no. 20-51-56017

Co-investigator on “Structural properties of random graphs and hypergraphs”. Russian Science Foundation, no. 22-21-00411

Teaching

TU Dortmund, Department of Computer Science

Complexity Theory — Design of study materials, tutoring

Higher School of Economics, Faculty of Computer Science

Theoretical Computer Science — Lecturer, course design

Mathematical Statistics — Design of study materials, tutoring

Algorithms and Data Structures — Design of study materials, tutoring

Theory of Computation — Tutoring

Convex Programming and Approximation Algorithms — Tutoring

Probability Theory — Tutoring

Service and outreach

ADYN Summerschool on Algorithms, Dynamics, and Information Flow in Networks

Organizing

“Combi meets Stat Physics” reading group, UNSW

Organizing

Peer reviewing and subreviewing

Annals of Applied Probability, Australasian Journal of Combinatorics,
Combinatorica, Random Structures and Algorithms, STACS

Project programme in mathematics & theoretical computer science, “Sirius”

2022

Educational Center

Mentor

Mathematics summer school “Kostroma open”

2017

Invited teacher

Other

Ilya Segalovich scholarship

2021-2022

Awarded by Yandex, €10K